ACTIVITY 1.1 · UNIT 1 — BUILDING THE BODY OF A ROBOT

Structures & Building

Student Edition — VEX EXP · C++ in VS Code

Name: ________________________________ **Date:** _____________ **Period:** _________

Introduction

Before a robot can move, sense, or run a program, it needs a frame — the **chassis** — to hold everything together. If the frame flexes, the wheels, arms, and motors mounted to it flex too. In this activity you'll meet the EXP building system and its two tools, learn why **triangulation** makes a structure rigid, then build a structure and brace it. Read Chapter 1 in the textbook for the full explanation.

Learning Objectives

- Identify the main VEX EXP structural parts (C-channels, flat bars, plates, gussets, standoffs) and the two kit tools.
- Explain why triangulation makes a structure rigid, and why a single-screw joint acts as a hinge.
- Build a stable structure and safely load-test it for flex, then brace it to reduce wobble.
- Practice organized, safe use of the kit and a parts-management routine.

Key Ideas (quick reference)

Parts: C-channels (strongest), flat bars, plates (for stiffening, not main structure), gussets (brace corners), standoffs, shafts/collars/bearings.

Tools: T15 Star Drive Key turns the #8-32 Star Drive Screw; the open-end wrench holds the #8-32 Low Profile Nut. *Wrench holds the nut, star key turns the screw.*

One screw is a hinge: a single-screw joint pivots — use two screws to lock an angle.

Triangulation: a diagonal turns a wobbly square into locked triangles. When you can't add a diagonal, a 45° gusset does the same job.

Equipment & Materials

- VEX EXP kit (structural metal, standoffs, shafts, fasteners)
- T15 Star Drive Key and open-end wrench
- A standard test load (identical water bottles, books, or a bucket + marbles)
- Engineering notebook

Procedure

1. **Inventory and sort** the kit; set up the parts-management routine you'll keep all semester.
2. **Practice the joins:** bolt a C-channel to a plate with a star screw and Low Profile Nut, using *two* screws so the joint can't pivot. Mount a shaft so it spins freely but is locked by a collar.
3. **Build a freestanding tower or bridge of your own design** (or one your teacher provides), then load-test it safely — note *where* it flexes most.
4. **Brace it.** Add a diagonal, a gusset, a standoff, or close a box where it wobbled. Re-test with the *same* load and record the before/after in your notebook.

Load-Test Safely

Test for **flex (wobble that springs back), not failure (permanent bending).** EXP metal is aluminum and can take a permanent bend if forced.

1. **Set a load cap before you start** and write it down. The test ends at the cap — that's not failure.
2. **No pushing down by hand, standing on it, or levering.** Force comes only from the agreed weights or a light finger-push.
3. **Measure deflection**; “pass” means it springs back. Reward the stiffest design, not a survived catastrophe.

Conclusion Questions

Answer in complete sentences.

1. Using triangulation, explain why a diagonal brace or a 45° gusset makes a structure stronger.

2. A C-channel and a flat bar are the same length. Which resists bending better, and what about its shape explains that?

3. You join two beams with one screw and they keep rotating. What's happening, and what's the fix?

4. Why does adding more screws to a joint make it stronger but isn't always the best choice?

5. Name one way to stiffen a drivetrain other than adding a diagonal brace.

6. In a load test, why do we measure how far the structure flexes instead of how much weight makes it break?